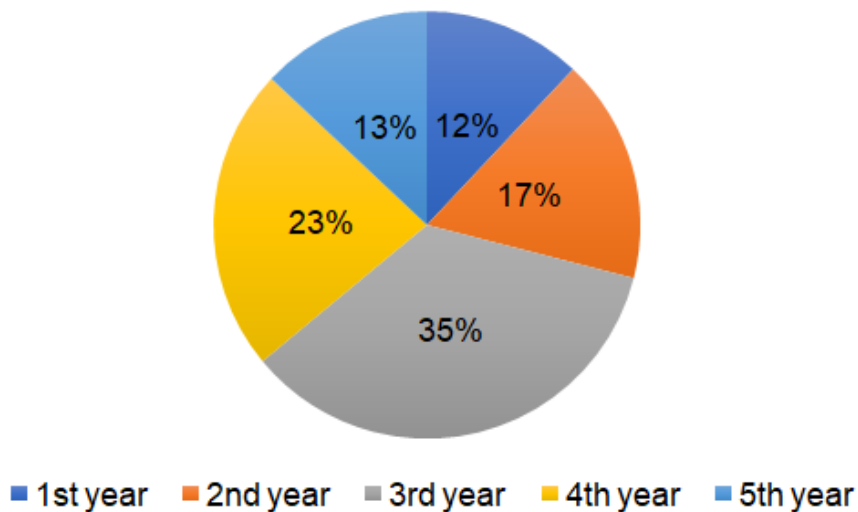


Directions: Answer the questions based on the information given below.

There are total 6400 books in a college library which started issuing books in the last week of November, out of which, 2400 books were issued to the students in the last week of November and 40% of the remaining books are issued in the following week on Monday. The pie chart given below shows the percentage distribution of the number of books issued on Tuesday to different types of students (1st year, 2nd year, 3rd year, 4th year and 5th year) of the college.

Note: Number of books issued to 2nd year students is 30 more than the number of books issued to 1st year students on Tuesday.

Percentage of books issued to different students on Tuesday



Question 1: If the number of books issued to 3rd year student on Monday is 50% more than the number of books issued to 3rd year student on Tuesday, find the number of books issued to 3rd year student on Monday.

- A) 315
- B) 325
- C) 295
- D) 335

E) None of these

Question 2: The number of books issued to 5th year students on Tuesday is what percentage of the number of books issued to 4th year students on the same day?

A) 52.4%

B) 56.5%

C) 61.3%

D) 65.9%

E) 49.1%

Question 3: Find the total number of books issued till Tuesday starting from the last week of November.

A) 4200

B) 4400

C) 4600

D) 4800

E) None of these

Question 4: 20% of the books issued till Tuesday was deposited on Wednesday, and 720 books were issued on Wednesday. Find the number of books left in the library on Wednesday at the end of the day.

A) 2000

B) 1900

C) 1800

D) 2100

E) None of these

Question 5: If each student of 5th year was issued 3 books on Tuesday, then find the number of 5th year students who were issued books on Tuesday.

A) 26

B) 28

C) 32

D) 36

E) None of these

Question 6: What approximate value will come in place of the question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

$$782.85^{1/2} + 37.505\% \text{ of } 2688.04 + 48.12 \times 34.25 \div (67.78) = ?$$

A) 780

B) 1340

C) 2160

D) 1060

E) 1680

Question 7: What approximate value will come in place of the question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

$$116.89 + 286.03 + 128.89 - 50\% \text{ of } 960.09 = 64.89\% \text{ of } ?$$

A) 420

- B) 680
- C) 240
- D) 580
- E) 80

Question 8: What approximate value will come in place of the question mark (?) in the following question?(Note: You are not expected to calculate the exact value.)

$$\{1244.9547 + 3.03 \times 1086.98 - (320.97 \times 11.0543)\} = ?$$

- A) 423
- B) 298
- C) 516
- D) 975
- E) 284

Question 9: What approximate value will come in place of the question mark (?) in the following question?(Note: You are not expected to calculate the exact value.)

$$(75.04\% \text{ of } 839.95 - 15.92 \times 15.13) = ? \times 26.09$$

- A) 95
- B) 15
- C) 135
- D) 65
- E) 185

Question 10: What approximate value will come in place of the question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

$$62.45\% \text{ of } 3200.09 + 40.12\% \text{ of } ? = 2095.94$$

- A) 240
- B) 120
- C) 380
- D) 20
- E) 520

Question 11: What approximate value will come in place of the question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

$$265.08 + (79.98\% \text{ of } 750.08) \div 15.12 + ? = 428.09$$

- A) 43
- B) 123
- C) 246`
- D) 372
- E) 423

Question 12: Mixture 'A' contains 80% acetone and rest 32 litres water. Mixture 'B' contains 72 litres acetone and rest 55% water. When two mixtures are mixed, then find the ratio of acetone to water in the resultant mixture.

- A) 2:1

B) 6:5

C) 5:3

D) 3:2

E) 4:3

Question 13: Ravish takes 32 days to complete a work while Arnab can complete the same work in 12 days less than Ravish. If Varun can complete 25% of the work in 12 days, then find the time taken by Ravish, Arnab and Varun together to complete 61.25% of the work.

A) 9 days

B) 4 days

C) 8 days

D) 12 days

E) 6 days

Question 14: Two dice were tossed simultaneously. Find the probability that the sum of the numbers obtained on the two dice is a prime number.

A) $\frac{5}{12}$

B) $\frac{3}{8}$

C) $\frac{4}{9}$

D) $\frac{1}{8}$

E) $\frac{2}{9}$

Question 15: A dog takes 5 leaps for every 4 leap of a frog. If 1 leap of dog is equal to 2 leaps of the frog, then find the ratio of the speed of the dog to that of the frog.

- A) 7:5
- B) 9:4
- C) 3:1
- D) 5:2
- E) 7:2

Question 16: Anderson is cycling at an average speed of 12 km/hr such that he can reach a certain point at 12 noon. If he cycles at 20 km/hr, then he will reach the destination at 10 a.m. At what speed he should cycle to reach the destination at 11 a.m.

- A) 15 km/hr
- B) 12.5 km/hr
- C) 16.5 km/hr
- D) 16 km/hr
- E) 14 km/hr

Question 17: The curved surface area of a cylinder whose radius of base and height are in the ratio 5:2, respectively, is 3080 cm^2 . If the circumference of the base of a cone is 132 cm less than that of the cylinder, then find the radius of the base of the cone.

- A) 7 cm
- B) 21 cm
- C) 35 cm
- D) 28 cm
- E) 14 cm

Question 18: The percentage profit earned by selling an article 'A' for Rs. 2800 is equal to the loss percent incurred by the selling the same article for Rs. 2200. Find the selling price when another article 'B' whose cost price is Rs. 700 more than that of article 'A' is sold at 15% profit.

- A) Rs. 3680
- B) Rs. 3240
- C) Rs. 3560
- D) Rs. 3450
- E) Rs. 3220

Question 19: When 12 years is subtracted from the present age of Vipul and the obtained result is divided by 3, then the present age of his nephew is obtained. If the present age of his nephew is 4 years less than the present age of Vipul's son who is 16 years old, then find the present age of Vipul.

- A) 36 years
- B) 54 years
- C) 48 years
- D) 33 years
- E) 51 years

Question 20: If the sum of the 1st 21 terms of an AP is equal to the sum of 1st 31 terms of that AP, then find the sum of the 1st 52 terms of the same AP.

- A) 0
- B) -212
- C) -325

D) 212

E) Can't be determined

Question 21: Suman sold an article at 40% profit. If the profit percentage had been numerically equal to CP, then the profit earned had been 250% more. Find the amount for which Suman sold the article originally.

A) Rs. 164

B) Rs. 208

C) Rs. 196

D) Rs. 224

E) None of these

Question 22: 240 metre long train 'A' is running with a speed of 72 km/hr. Train 'B' which is 360 metre long is running with a speed of 108 km/h in opposite direction of train 'A'. For how much time, the smaller train is completely obscured by the larger train?

A) 2.4 seconds

B) 5.6 seconds

C) 3.2 seconds

D) 1.8 seconds

E) None of these

Question 23: There are total 300 students in three sections A, B and C of a class. The average marks of students all three sections were 90.7. The average marks of students of section B and C was 92.5 and the average marks of students of section A was 88. Ratio of number of students in section B and C is 5:4, respectively. Find number of students in section B.

- A) 80
- B) 100
- C) 120
- D) 60
- E) None of these

Question 24: A train can cross 148 metres long platform in 20.5 seconds while a 192.8 metres long tunnel in 23.3 seconds. Find the time taken by the train to cross a man who is coming towards it at a rate of 28.8 km/h

- A) 7.5 seconds
- B) 6.5 seconds
- C) 9.5 seconds
- D) 8.5 seconds
- E) 12.5 seconds

Question 25: The following numbers form a series. Find the odd one out.

114, 138, 98, 154, 82, 172

- A) 138
- B) 98
- C) 82
- D) 114
- E) 172

Question 26: The following numbers form a series. Find the odd one out.

8, 19, 61, 249, 1251, 7517

- A) 19
- B) 7517
- C) 249
- D) 1251
- E) 61

Question 27: The following numbers form a series. Find the odd one out.

13, 30, 128, 147, 588, 607

- A) 30
- B) 607
- C) 128
- D) 147
- E) 588

Question 28: The following numbers form a series. Find the odd one out.

228, 307, 188, 357, 132, 421

- A) 421
- B) 132
- C) 188
- D) 357
- E) 307

Question 29: The following numbers form a series. Find the odd one out.

13, 21, 32, 46, 63, 85

A) 21

B) 85

C) 46

D) 63

E) 32

Question 30: The following numbers form a series. Find the odd one out.

132, 148, 196, 276, 378, 532

A) 276

B) 532

C) 378

D) 148

E) 196

Direction: Answer the questions based on the information given below.

A shopkeeper sold two types of shawls i.e. Pashmina and Knit on different days of a week. The table given below shows the total number of shawls sold, and ratio of number of Pashmina to Knit shawls sold by the shopkeeper.

Days	Total number of shawls sold	Ratio of number of Pashmina to Knit shawls sold
Monday	1200	8:7

Tuesday	1440	7:5
Thursday	1080	5:4
Wednesday	1280	9:7
Friday	1620	4:5

Question 31: What is the ratio of number of Pashmina shawls sold on Monday to number of Knit shawls sold on Thursday?

- A) 2:3
- B) 4:3
- C) 3:1
- D) 5:4
- E) None of these

Question 32: What is the average of number of shawls sold by the shopkeeper on the given days except Thursday?

- A) 1395
- B) 1375
- C) 1385
- D) 1345
- E) None of these

Question 33: Number of Pashmina shawls sold on Tuesday is how much percent more/less than the number of Knit shawls sold on Wednesday?

- A) 70%

- B) 50%
- C) 75%
- D) 60%
- E) None of these

Question 34: What is the difference between number of Pashmina shawls sold on Thursday, and number Knit shawls sold on Friday?

- A) 280
- B) 320
- C) 300
- D) 400
- E) None of these

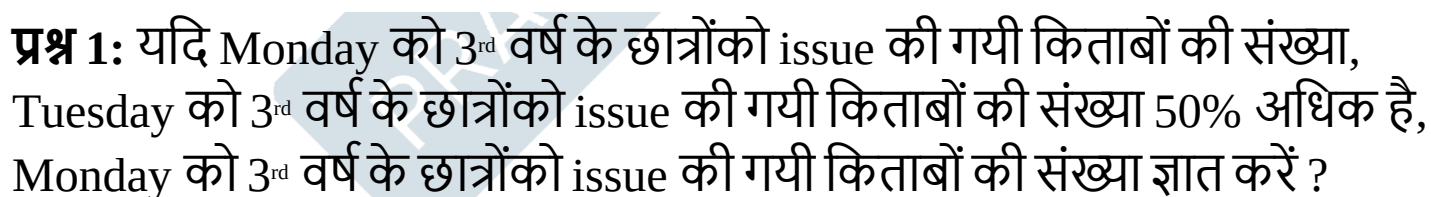
Question 35: Number of Pashmina and Knit shawls sold on Saturday is 25% more and 20% more, respectively than the same sold on Friday and Tuesday. Total number of shawls sold on Saturday is:

- A) 1600
- B) 1640
- C) 1480
- D) 1520
- E) None of these

निर्देश: नीचे दी गई जानकारी के आधार पर प्रश्नों के उत्तर दें।

एक college के पुस्तकालय में कुल 6400 पुस्तकें हैं, जिसने November के अंतिम सप्ताह में किताबें issue करना शुरू किया था, जिसमें से 2400 पुस्तकें November

नोट: Tuesday को 2nd वर्ष के छात्रों को जारी की गई पुस्तकों की संख्या Tuesday को 1st वर्ष, के छात्रों के लिए issue की गई पुस्तकों की संख्या से 30 अधिक है।



- A) 315
- B) 325
- C) 295
- D) 335
- E) \u0907\u0928\u092e\u0947\u0902 \u0938\u0947 \u0915\u094d\u0902\u0924\u0940\u0902

प्रश्न 2: Tuesday को 5th वर्ष के छात्रों को issue की गयी किताबों की संख्या, उसी दिन 4th वर्ष के छात्रों को issue की गयी किताबों की संख्या का कितना प्रतिशत है ?

- A) 52.4%
- B) 56.5%
- C) 61.3%
- D) 65.9%
- E) 49.1%

प्रश्न 3: November के अंतिम सप्ताह से शुरू होते हुए Tuesday तक issue पुस्तकों की कुल संख्या ज्ञात करें ?

- A) 4200
- B) 4400
- C) 4600
- D) 4800
- E) \u007\u0028\u002e\u0047\u0002 \u0038\u0047 \u0015\u004b\u0008 \u0028\u0039\u0040

प्रश्न 4: Tuesday तक issued 20% किताबें Wednesday, को जमा की गईं, और Wednesday को 720 किताबें issue की गईं। दिन के अंत में Wednesday को पुस्तकालय में बची हुई पुस्तकों की संख्या ज्ञात करें ?

- A) 2000
- B) 1900
- C) 1800
- D) 2100

- B) 680
- C) 240
- D) 580
- E) 80

प्रश्न 8: निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या अनुमानित मान आएगा?
(नोट: आपको सटीक मान की गणना करने की आवश्यकता नहीं है।)

$$\{1244.9547 + 3.03 \times 1086.98 - (320.97 \times 11.0543)\} = ?$$

- A) 423
- B) 298
- C) 516
- D) 975
- E) 284

प्रश्न 9: निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या अनुमानित मान आएगा?
(नोट: आपको सटीक मान की गणना करने की आवश्यकता नहीं है।)

$$(75.04\% \text{ of } 839.95 - 15.92 \times 15.13) = ? \times 26.09$$

- A) 95
- B) 15
- C) 135
- D) 65
- E) 185

प्रश्न 10: निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या अनुमानित मान आएगा? (नोट: आपको सटीक मान की गणना करने की आवश्यकता नहीं है।)

$$62.45\% \text{ of } 3200.09 + 40.12\% \text{ of } ? = 2095.94$$

- A) 240
- B) 120
- C) 380
- D) 20
- E) 520

प्रश्न 11: निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या अनुमानित मान आएगा? (नोट: आपको सटीक मान की गणना करने की आवश्यकता नहीं है।)

$$265.08 + (79.98\% \text{ of } 750.08) \div 15.12 + ? = 428.09$$

- A) 43
- B) 123
- C) 246
- D) 372
- E) 423

प्रश्न 12: मिश्रण 'A' में 80% एसीटोन और शेष 32 litres पानी है। मिश्रण 'B' में 72 litres एसीटोन और शेष 55% पानी है। जब दो मिश्रणों को मिलाया जाता है, तो परिणामी मिश्रण में एसीटोन और पानी का अनुपात ज्ञात कीजिए।

- A) 2:1
- B) 6:5

C) 5:3

D) 3:2

E) 4:3

प्रश्न 13: Ravish एक काम को पूरा करने में 32 दिन लेता है जबकि Arnab उसी काम को Ravish से 12 दिन कम में पूरा कर सकता है। यदि Varun काम का 25% 12 दिनों में पूरा कर सकता है, तो Ravish, Arnab और Varun द्वारा साथ में काम का 61.25% पूरा करने में लगने वाला समय ज्ञात कीजिए।

A) 9 days

B) 4 days

C) 8 days

D) 12 days

E) 6 days

प्रश्न 14: दो पासे एक साथ उछाले गए। दोनों पासों पर प्राप्त संख्याओं का योग एक अभाज्य संख्या होने की प्रायिकता ज्ञात कीजिए।

A) 5/12

B) 3/8

C) 4/9

D) 1/8

E) 2/9

प्रश्न 15: एक कुत्ता मेंढक की हर 4 छलांग पर 5 छलांग लगाता है। यदि कुत्ते की 1 छलांग मेंढक की 2 छलांग के बराबर है, तो कुत्ते की गति और मेंढक की गति का अनुपात ज्ञात कीजिए।

- A) 7:5
- B) 9:4
- C) 3:1
- D) 5:2
- E) 7:2

प्रश्न 16: Anderson 12 km/hr की औसत गति से साइकिल चला रहा है ताकि वह दोपहर 12 बजे एक निश्चित बिंदु तक पहुंच सके। यदि वह 20 km/hr की गति से साइकिल चलाता है, तो वह सुबह 10 बजे गंतव्य पर पहुंच जाएगा। 11 बजे गंतव्य तक पहुंचने के लिए उसे कितनी गति से साइकिल चलानी चाहिए।

- A) 15 km/hr
- B) 12.5 km/hr
- C) 16.5 km/hr
- D) 16 km/hr
- E) 14 km/hr

प्रश्न 17: एक बेलन का वक्र पृष्ठीय क्षेत्रफल, जिसकी आधार की त्रिज्या और ऊँचाई क्रमशः 5:2 के अनुपात में है, 3080 cm^2 है। यदि एक शंकु के आधार की परिधि बेलन की तुलना में 132 cm कम है, तो शंकु के आधार की त्रिज्या ज्ञात कीजिए।

- A) 7 cm
- B) 21 cm
- C) 35 cm
- D) 28 cm
- E) 14 cm

प्रश्न 18: एक वस्तु 'A' को Rs. 2800 में बेचकर अर्जित प्रतिशत लाभ उसी वस्तु को Rs. 2200 में बेचने पर हुई हानि प्रतिशत के बराबर है। विक्रय मूल्य ज्ञात कीजिए जब एक अन्य वस्तु 'B', जिसका क्रय मूल्य वस्तु 'A' से Rs. 700 अधिक है, को 15% लाभ पर बेचा जाता है।

- A) Rs. 3680
- B) Rs. 3240
- C) Rs. 3560
- D) Rs. 3450
- E) Rs. 3220

प्रश्न 19: जब Vipul की वर्तमान आयु में से 12 वर्ष घटाया जाता है और प्राप्त परिणाम को 3 से विभाजित किया जाता है, तो उसके भतीजे की वर्तमान आयु प्राप्त होती है। यदि उसके भतीजे की वर्तमान आयु Vipul के पुत्र की वर्तमान आयु से 4 वर्ष कम है जो 16 वर्ष का है, तो Vipul की वर्तमान आयु ज्ञात कीजिए।

- A) 36 years
- B) 54 years
- C) 48 years
- D) 33 years
- E) 51 years

प्रश्न 20: यदि किसी अंकगणितीय श्रेढ़ी के पहले 21 पदों का योग उस अंकगणितीय श्रेढ़ी के पहले 31 पदों के योग के बराबर है, तो उसी अंकगणितीय श्रेढ़ी के पहले 52 पदों का योग ज्ञात कीजिए।

- A) 0
- B) -212

C) -325

D) 212

E) निर्धारित नहीं किया जा सकता

प्रश्न 21: Suman ने 40% लाभ पर एक वस्तु को बेचा। यदि लाभ प्रतिशत संख्यात्मक रूप से क्रय मूल्य के बराबर होता तो अर्जित लाभ 250% अधिक होता। वह राशि ज्ञात करें जिसके लिए Suman ने वास्तविक रूप से वस्तु को बेचा था।

A) Rs. 164

B) Rs. 208

C) Rs. 196

D) Rs. 224

E) इनमें से कोई नहीं

प्रश्न 22: 240 metre लंबी ट्रेन 'A' 72 km/hr की गति से चल रही है। ट्रेन 'B' जो कि 360 metre लंबी है, ट्रेन 'A' के विपरीत दिशा में 108 km/h की गति से चल रही है। कितने समय के लिए, बड़ी ट्रेन से छोटी ट्रेन पूरी तरह से ढक जाती है?

A) 2.4 seconds

B) 5.6 seconds

C) 3.2 seconds

D) 1.8 seconds

E) इनमें से कोई नहीं

प्रश्न 23: एक कक्षा के तीन sections A, B और C में कुल 300 छात्र हैं। तीनों sections के छात्रों के औसत अंक 90.7 थे। section B और C के छात्रों का औसत अंक 92.5 था और section A के छात्रों का औसत अंक 88 था। Section B और C

में छात्रों की संख्या का अनुपात क्रमशः 5: 4 है। Section B में छात्रों की संख्या ज्ञात करें?

- A) 80
- B) 100
- C) 120
- D) 60
- E) इनमें से कोई नहीं

प्रश्न 24: एक ट्रेन 20.5 seconds में 148 metres लंबे प्लेटफार्म को जबकि 192.8 metres लंबी सुरंग को 23.3 seconds में पार कर सकती है। उस व्यक्ति को पार करने में ट्रेन द्वारा लगने वाले समय ज्ञात करें जो 28.8 km/h की दर से ट्रेन की ओर आ रहा है।

- A) 7.5 seconds
- B) 6.5 seconds
- C) 9.5 seconds
- D) 8.5 seconds
- E) 12.5 seconds

प्रश्न 25: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

114, 138, 98, 154, 82, 172

- A) 138
- B) 98
- C) 82
- D) 114

E) 172

प्रश्न 26: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

8, 19, 61, 249, 1251, 7517

A) 19

B) 7517

C) 249

D) 1251

E) 61

प्रश्न 27: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

13, 30, 128, 147, 588, 607

A) 30

B) 607

C) 128

D) 147

E) 588

प्रश्न 28: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

228, 307, 188, 357, 132, 421

A) 421

B) 132

C) 188

D) 357

E) 307

प्रश्न 29: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

13, 21, 32, 46, 63, 85

A) 21

B) 85

C) 46

D) 63

E) 32

प्रश्न 30: निम्नलिखित संख्या एक श्रृंखला बनाती है। विषम का चुनाव करें?

132, 148, 196, 276, 378, 532

A) 276

B) 532

C) 378

D) 148

E) 196

निर्देश: नीचे दी गई जानकारी के आधार पर प्रश्नों के उत्तर दें।

एक दुकानदार सप्ताह के विभिन्न दिनों में दो प्रकार के shawls बेचता है यानि Pashmina और Knit। नीचे दी गई तालिका में दुकानदार द्वारा बेची गई shawls की कुल संख्या, और बेचे गए Pashmina और Knit shawls की संख्या का अनुपात दिखाया गया है।

Days	Total number of shawls sold	Ratio of number of Pashmina to Knit shawls sold
Monday	1200	8:7
Tuesday	1440	7:5
Thursday	1080	5:4
Wednesday	1280	9:7
Friday	1620	4:5

प्रश्न 31: Monday को बेची गई Pashmina shawls की संख्या और Thursday को बेची गई Knit shawls की संख्या का अनुपात ज्ञात करें?

- A) 2:3
- B) 4:3
- C) 3:1
- D) 5:4
- E) इनमें से कोई नहीं

प्रश्न 32: Thursday को छोड़कर दिए गए दिनों में दुकानदार द्वारा बेचे गए shawls की संख्या का औसत कितना है?

- A) 1395
- B) 1375
- C) 1385
- D) 1345
- E) इनमें से कोई नहीं

प्रश्न 33: Tuesday को बेची गई Pashmina shawls की संख्या Wednesday को बेची गई Knit shawls की संख्या से कितने प्रतिशत अधिक / कम है?

- A) 70%
- B) 50%
- C) 75%
- D) 60%
- E) इनमें से कोई नहीं

प्रश्न 34: Thursday को बेची गई Pashmina shawls की संख्या और Friday को बेचे जाने वाले Knit shawls के बीच का अंतर ज्ञात करें?

- A) 280
- B) 320
- C) 300
- D) 400
- E) इनमें से कोई नहीं

प्रश्न 35: Saturday को बेची गई Pashmina और Knit shawls की संख्या Friday और Tuesday को बेची गई Pashmina और Knit shawls की संख्या से 25% अधिक और 20% अधिक है। Saturday को बेचे गए shawls की कुल संख्या है:

- A) 1600
- B) 1640
- C) 1480
- D) 1520
- E) इनमें से कोई नहीं

ANSWER KEYS and SOLUTIONS:

1) - A)	2) - B)	3) - C)	4) - A)	5) - A)	6) - D)
7) - E)	8) - D)	9) - B)	10) - A)	11) - B)	12) - C)
13) - E)	14) - A)	15) - D)	16) - A)	17) - E)	18) - A)
19) - C)	20) - A)	21) - C)	22) - A)	23) - B)	24) - A)
25) - E)	26) - B)	27) - A)	28) - E)	29) - B)	30) - C)
31) - B)	32) - C)	33) - B)	34) - C)	35) - E)	

Solution 1: A)

Let the total number of books issued on Tuesday be 'x'.

$$\text{So, } (17 - 12)\% \text{ of } x = 30$$

$$5\% \text{ of } x = 30$$

$$x = 600$$

Number of books issued to 3rd year student on Tuesday = 35% of 600 = 210

Number of books issued to 3rd year student on Monday = $210 \times 1.5 = 315$

Hence, option a.

Solution 2: B)

Let the total number of books issued on Tuesday be 'x'.

$$\text{So, } (17 - 12)\% \text{ of } x = 30$$

$$5\% \text{ of } x = 30$$

$$x = 600$$

Number of books issued to 4th year student on Tuesday = 23% of 600 = 138

Number of books issued to 5th year student on Tuesday = 13% of 600 = 78

Required percentage = $78/138 \times 100 \sim 56.5\%$

Hence, option b.

Solution 3: C)

Let the total number of books issued on Tuesday be 'x'.

$$\text{So, } (17 - 12)\% \text{ of } x = 30$$

$$5\% \text{ of } x = 30$$

$$x = 600$$

Number of books already issued = 2400

Number of books issued on Monday = 40% of $(6400 - 2400) = 1600$

Number of books issued on Tuesday = 600

So, number of books issued till Tuesday = $2400 + 1600 + 600 = 4600$

Hence, option c.

Solution 4: A)

Let the total number of books issued on Tuesday be 'x'.

$$\text{So, } (17 - 12)\% \text{ of } x = 30$$

$$5\% \text{ of } x = 30$$

$$x = 600$$

Number of books already issued = 2400

Number of books issued on Monday = 40% of $(6400 - 2400) = 1600$

Number of books issued on Tuesday = 600

Number of books deposited till Tuesday = 20% of $4600 = 920$

Number of books left in the library = $6400 - 4600 + 920 - 720 = 2000$

Hence, option a.

Solution 5: A)

Let the total number of books issued on Tuesday be 'x'.

$$\text{So, } (17 - 12)\% \text{ of } x = 30$$

$$5\% \text{ of } x = 30$$

$$x = 600$$

Number of books issued to 5th year student on Tuesday = 13% of 600 = 78

Number of 5th year students who were issued books on Tuesday = $78/3 = 26$

Hence, option a.

Solution 6: D)

$$782.85^{1/2} + 37.505\% \text{ of } 2688.04 + 48.12 \times 34.25 \div (67.78) = ?$$

$$? \sim 784^{1/2} + 37.5\% \text{ of } 2688 + 48 \times 34 \div 68$$

$$? \sim 28 + 1008 + 24$$

$$? \sim 1060$$

Hence, option d.

Solution 7: E)

$$116.89 + 286.03 + 128.89 - 50\% \text{ of } 960.09 = 64.89\% \text{ of } ?$$

$$65\% \text{ of } ? \sim 117 + 286 + 129 - 50\% \text{ of } 960$$

$$65\% \text{ of } ? \sim 532 - 480$$

$$65\% \text{ of } ? \sim 52$$

$$(13/20) \times ? \sim 52$$

$$? \sim 80$$

Hence, option e.

Solution 8: D)

$$\{1244.9547 + 3.03 \times 1086.98 - (320.97 \times 11.0543)\} = ?$$

$$\{1245 + 3 \times 1087 - (321 \times 11)\} \sim ?$$

$$(1245 + 3261 - 3531) \sim ?$$

$$? \sim 975$$

Hence, option d.

Solution 9: B)

$$(75.04\% \text{ of } 839.95 - 15.92 \times 15.13) = ? \times 26.09$$

$$(75\% \text{ of } 840 - 16 \times 15) \sim ? \times 26$$

$$(630 - 240) = ? \times 26$$

$$390 = ? \times 26$$

$$? \sim 15$$

Hence, option b.

Solution 10: A)

$$62.45\% \text{ of } 3200.09 + 40.12\% \text{ of } ? = 2095.94$$

$$62.5\% \text{ of } 3200 + 40\% \text{ of } ? \sim 2096$$

$$0.625 \times 3200 + 0.4 \times ? \sim 2096$$

$$2000 + 0.4 \times ? \sim 2096$$

$$0.4 \times ? \sim 96$$

$$? \sim 240$$

Hence, option a.

Solution 11: B)

$$265.08 + (79.98\% \text{ of } 750.08) \div 15.12 + ? = 428.09$$

$$265 + (80\% \text{ of } 750) \div 15 + ? \sim 428$$

$$265 + 600 \div 15 + ? \sim 428$$

$$265 + 40 + ? \sim 428$$

$$305 + ? \sim 428$$

$$? \sim 123$$

Hence, option b.

Solution 12: C)

Quantity of water in mixture 'A' = 32 litres

Total quantity of mixture 'A' = $32/0.2 = 160$ litres

Therefore, quantity of acetone in mixture 'A' = $160 - 32 = 128$ litres

Quantity of acetone in mixture 'B' = 72 litres

Total quantity of mixture 'B' = $72/0.45 = 160$ litres

Quantity of water in mixture 'B' = $160 - 72 = 88$ litres

Therefore, quantity of acetone in resultant mixture = $128 + 72 = 200$ litres

Quantity of water in resultant mixture = $32 + 88 = 120$ litres

Required ratio = $200:120 = 5:3$

Hence, option c.

Solution 13: E)

Time taken by Ravish to complete the work = 32 days

Time taken by Arnab to complete the work = $32 - 12 = 20$ days

Time taken by Varun to complete the work = $12/0.25 = 48$ days

Let the total work = 480 units

Efficiency of Ravish = $480/32 = 15$ units/day

Efficiency of Arnab = $480/20 = 24$ units/day

Efficiency of Varun = $480/48 = 10$ units/day

Therefore, 1 days work of (Varun + Ravish + Arnab) = $15 + 24 + 10 = 49$ units

Therefore, required time taken = $0.6125 \times 480/49 = 6$ days

Hence, option e.

Solution 14: A)

Total number of outcomes = $6 \times 6 = 36$

Events of getting a sum which is a prime number = $\{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\} = 15$

Required probability = Number of favourable outcomes/total number of outcomes

Or, required probability = $15/36 = 5/12$

Hence, option a.

Solution 15: D)

According to the question,

1 leap of dog = 2 leaps of frog

Therefore, 5 leap of dog = $5 \times 2 = 10$ leaps of the frog

Therefore, ratio of speed of the dog to the speed of the frog = $10:4 = 5:2$

Hence, option d.

Solution 16: A)

Let the distance to be travelled be 'x' km

Time taken when speed is 12 km/hr = $(x/12)$ hours

Time taken when speed is 20 km/hr = $(x/20)$ hours

According to the question,

$$(x/12) - (x/20) = 2$$

$$\text{Or, } 5x - 3x = 2 \times 60$$

$$\text{Or, } 2x = 120$$

$$\text{Or, } x = 60 \text{ km}$$

Time taken to travel 60 km with speed of 12 km/hr = $60/12 = 5$ hours

Therefore, he started at $12 - 5 = 7$ a.m.

Therefore, required speed to reach the destination at 11 a.m. or 4 hours = $60/4 = 15$ km/hr

Hence, option a.

Solution 17: E)

Let the radius of the base and the height of the cylinder be $5x$ cm and $2x$ cm, respectively

According to the question,

$$2\pi rh = 3080$$

$$2 \times (22/7) \times 5x \times 2x = 3080$$

$$\text{Or, } x^2 = 3080 \times (7/440) = 49$$

$$\text{Or, } x = 7$$

Therefore, radius of the base of the cylinder = $5x = 35$ cm

Circumference of the base of the cone = $\{2 \times (22/7) \times 35\} - 132 = 88$ cm

Let the radius of the base of the cone be ' r ' cm

$$\text{Therefore, } 2\pi r = 88$$

$$\text{Or, } 2 \times (22/7) \times r = 88$$

$$\text{Or, } r = (88/44) \times 7$$

$$\text{Or, } r = 14 \text{ cm}$$

Hence, option e.

Solution 18: A)

Let the cost price of the article 'A' be Rs. x

Therefore, profit percent = $\{(2800 - x)/x\} \times 100$

And, loss percent incurred = $\{(x - 2200)/x\} \times 100$

According to the question,

$$\{(2800 - x)/x\} \times 100 = \{(x - 2200)/x\} \times 100$$

$$\text{Or, } 2800 - x = x - 2200$$

$$\text{Or, } 2x = 5000$$

$$\text{Or, } x = 2500$$

Therefore, cost price of article 'B' = $2500 + 700 = \text{Rs. } 3200$

Therefore, required selling price of article 'B' = $1.15 \times 3200 = \text{Rs. } 3680$

Hence, option a.

Solution 19: C)

Let the Present age of Vipul be 'x' years

Present age of his nephew = $16 - 4 = 12$ years

Therefore,

$$\{(x - 12)/3\} = 12$$

$$\text{Or, } x - 12 = 36$$

$$\text{Or, } x = 48$$

Therefore, present age of Vipul = 48 years

Hence, option c.

Solution 20: A)

Let the 1st term and common difference of the AP be 'a' and 'd', respectively

Sum of n terms i.e. S_n of an AP = $(n/2)\{2a + (n - 1)d\}$

$$S_{21} = (21/2)(2a + 20d)$$

$$S_{31} = (31/2)(2a + 30d)$$

According to the question,

$$(21/2)(2a + 20d) = (31/2)(2a + 30d)$$

$$\text{Or, } 42a + 420d = 62a + 930d$$

$$\text{Or, } 20a = -510d$$

$$\text{Or, } a = -25.5d$$

$$\text{Therefore, sum of 1}^{\text{st}} 52 \text{ terms of the AP} = (52/2)\{2 \times (-25.5)d + (52 - 1)d\} = 26(-51d + 51d) = 0$$

Hence, option a.

Solution 21: C)

Let the cost price of the article be Rs. $100x$

Therefore, selling price of article = Rs. $140x$

Therefore, profit earned = $140x - 100x = \text{Rs. } 40x$

New profit = $3.5 \times 40x = \text{Rs. } 140x$

According to the question,

$$(140x/100x) \times 100 = 100x$$

$$\text{Or, } x = 1.4$$

Therefore, amount for which Suman sold the article, originally = $140x = \text{Rs. } 196$

Hence, option c.

Solution 22: A)

Length of smaller train = 240 metre

Speed of smaller train = $72 \times (5/18) = 20$ m/s

Length of larger train = 360 metres

Speed of larger train = $108 \times (5/18) = 30$ m/s

Since, the smaller train is completely obscured by larger train, therefore,
extra length of larger train = $360 - 240 = 120$ metres

Relative speed of trains 'A' and 'B' = $30 + 20 = 50$ m/s

Required time = $120/50 = 2.4$ seconds

Hence, option a.

Solution 23: B)

Let number of students in section A be 'x'

Number of students in section B and C together = $300 - x$

According to question;

$$88x + 92.5 \times (300 - x) = 90.7 \times 300$$

$$88x + 27750 - 92.5x = 27210$$

$$4.5x = 540$$

$$x = 120$$

Number of students in sections B and C together = $300 - 120 = 180$

Number of students in section B = $5/9 \times 180 = 100$

Hence, option b.

Solution 24: A)

Let speed of train is 's' km/h and length of train is 'l' metres.

$$\text{So, } l + 148 = 20.5s \dots\dots\dots(1)$$

$$\text{And, } l + 192.8 = 23.3s \dots\dots\dots(2)$$

Equation (2) – Equation (1), we get

$$44.8 = 2.8s$$

$$\text{Or, } s = 16$$

$$\text{Length of train} = l = 20.5 \times 16 - 148 = 180 \text{ metres}$$

$$\text{Speed of man} = 28.8 \times 5/18 = 8 \text{ m/s}$$

$$\text{Desired time} = 180/(16 + 8) = 7.5 \text{ seconds}$$

Hence, option a.

Solution 25: E)

$$114 + 8 \times 3 = 138$$

$$138 - 8 \times 5 = 98$$

$$98 + 8 \times 7 = 154$$

$$154 - 8 \times 9 = 82$$

$$\mathbf{82 + 8 \times 11 = 170}$$

Therefore, 170 should come in place of 172.

Hence, option e.

Solution 26: B)

$$8 \times 2 + 3 = 19$$

$$19 \times 3 + 4 = 61$$

$$61 \times 4 + 5 = 249$$

$$249 \times 5 + 6 = 1251$$

$$1251 \times 6 + 7 = 7513$$

Therefore, 7513 should come in place of 7517.

Hence, option b.

Solution 27: A)

$$13 + 19 = 32$$

$$32 \times 4 = 128$$

$$128 + 19 = 147$$

$$147 \times 4 = 588$$

$$588 + 19 = 607$$

Therefore, 32 should come in place of 30.

Hence, option a.

Solution 28: E)

$$228 + 9^2 = 309$$

$$309 - 11^2 = 188$$

$$188 + 13^2 = 357$$

$$357 - 15^2 = 132$$

$$132 + 17^2 = 421$$

Therefore, 309 should come in place of 307.

Hence, option e.

Solution 29: B)

$$13 + 8 = 21$$

$$21 + 11 = 32$$

$$32 + 14 = 46$$

$$46 + 17 = 63$$

$$63 + 20 = 83$$

Therefore, 83 should come in place of 85.

Hence, option b.

Solution 30: C)

$$132 + 16 \times 1 = 148$$

$$148 + 16 \times 3 = 196$$

$$196 + 16 \times 5 = 276$$

$$276 + 16 \times 7 = 388$$

$$388 + 16 \times 9 = 532$$

So, 388 should come in place of 378.

Hence, option c.

Solution 31: B)

Days	Number of Pashmina shawls sold	Number of Knit shawls sold
Monday	$8/15 \times 1200 = 640$	$7/15 \times 1200 = 560$
Tuesday	$7/12 \times 1440 = 840$	$5/12 \times 1440 = 600$
Wednesday	$9/16 \times 1280 = 720$	$7/16 \times 1280 = 560$
Thursday	$5/9 \times 1080 = 600$	$4/9 \times 1080 = 480$
Friday	$4/9 \times 1620 = 720$	$5/9 \times 1620 = 900$

Desired ratio = $640:480 = 4:3$

Hence, option b.

Solution 32: C)

Desired Average = $(1200 + 1440 + 1280 + 1620)/4 = 1385$

Hence, option c.

Solution 33: B)

Desired Percentage = $[(840 - 560)/560] \times 100 = 50\%$

Hence, option b.

Solution 34: C)

Desired difference = $900 - 600 = 300$

Hence, option c.

Solution 35: E)

Total number of shawls sold on Saturday = $1.25 \times 720 + 1.20 \times 600 = 900 + 720 = 1620$

Hence, option e.